

Programming Using C++ Level 1

Logical Operators



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Definition

◆ Logical operators are used in C++ to combine or reverse Boolean conditions.

 The main logical operators are:

 `&&` → AND

 `||` → OR

 `!` → NOT

Explanation

- ◆ This program uses logical operators to evaluate multiple Boolean conditions and stores the final result in the Boolean variable Result.
- ◆ && returns True when both conditions are True.
- ◆ || returns True when at least one condition is True.
- ◆ ! reverses the Boolean result. True becomes False, and False becomes True.

Steps in main()

1. Declare the Boolean variable

◆ Create a Boolean variable called Result.

2. Evaluate the first expression

◆ Calculate:

Result = !(5 > 6 || 7 == 7) && !(1 || 0);

3. Print the first result

◆ Display the value stored in Result.

4. Evaluate the second expression

◆ Calculate:

Result = (7 == 7) && (7 > 5);

5. Print the second result

◆ Display the new value stored in Result.

💡 Example

◆ For:

A = True

B = False

▣ A && B → False

▣ A || B → True

▣ !A → False

Note

- ◆ 1 represents True and 0 represents False when used as Boolean values.
- ◆ The bool data type is used to store Boolean values.

⊘ Warning

⚡ Do not confuse `&&` with `||`.

Comparison

&& AND

- ✓ Both conditions must be True.
- ✓ True && True → True
- ✓ Otherwise → False

|| OR

- ✓ At least one condition must be True.
- ✓ False || False → False
- ✓ Otherwise → True

! NOT

- ✓ Reverses the Boolean value.
- ✓ True becomes False.
- ✓ False becomes True.

✦✦ Summary

- ✓ && means AND, || means OR, and ! means NOT.
- ✓ Logical operators are used to combine or reverse Boolean conditions.
- ✓ The result of a logical expression is True or False, represented by 1 or 0.

Thank you 😊